Bias in AI-to-AI loops (working draft)

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Summary

A caption-to-image-to-caption loop forgets who it was asked to picture within a few steps, and what takes the person's place is the loop's own default: light-skinned, male, and soon not a single person at all. Seeded with 36 occupation prompts that varied age, gender and skin tone, the loop had lost the skin tone of every dark-skinned seed by step 5 and all three attributes of every occupation seed by step 10. The replacement was not the occupation's stereotype. A dark-skinned social worker, whose one-shot rendering is dark-skinned 50% of the time, lost her skin tone as fast as a dark-skinned pilot. The bias that shows once the seed is gone belongs to the loop, and it is carried by the captioner.

This is a working draft by an independent AI experimenter, and it is explicitly a request for help. I ran the experiments with Claude Code as engineer and analyst, pre-registered predictions before each run, and hand-labelled a check set to calibrate the automated judge. I am not trained in bias research, several of the measurement choices below were made by two amateurs, and the experiments that should come next need someone who knows the field to design them. Corrections, reanalysis and collaboration are welcome.

Six findings:

1. **Seeded demographics are lost within a few steps.** Skin tone first, at step 1; gender and age over about ten.
2. **The replacement is the loop's prior, not the occupation's.** Faces end at the lightness of the model's neutral "a person", lighter than any occupation's own one-shot faces.
3. **The captioner is the channel.** Most losses are omissions. When the captioner does name a skin tone, it often names a dark face "fair". One occupation, social worker, is never named at all, so its stereotype never enters the loop.
4. **Asking the captioner to track attributes helps gender and age, not skin, and creates a new attractor**: grids of faces of "various ages, genders and skin tones".
5. **The loop re-assigns occupations, and the re-assignment is where the stereotype shows.** A nurse becomes a doctor when the picture has drifted to a light-skinned man. A woman pilot becomes a male astronaut.

6. **Whose default depends on the captioner and the language.** Gendered vocabulary ends male under one captioner and female under two others; a Chinese instruction turns faces East Asian within five steps.

Background

Text-to-image models carry demographic stereotypes, and this is well documented for a single prompt. Ghosh and Caliskan showed that Stable Diffusion's default person is a light-skinned Western man and that women of colour are sexualised in its outputs (Ghosh and Caliskan, 2023). Occupation prompts return the stereotyped gender and skin tone at rates far above workforce statistics. Wilson, Ghosh and Caliskan found that Stable Diffusion XL, the generator used here, ties stigmatised identities to specific skin tones and renders about 30% less skin-tone variability than earlier versions. Those are one-shot measurements: one prompt, one distribution of images.

An AI-to-AI loop is different. A caption is rendered, the image is described by a vision-language model, and the description becomes the next prompt. Hintze and colleagues showed that such loops converge to generic motifs within about twenty steps (Hintze et al., 2025). A companion draft measures that convergence as attractor dynamics. The question here is what the convergence does to the people in the pictures: whether a loop seeded with a counter-stereotypical person keeps them, loses them to the stereotype, or loses them to something else.

One group has asked a version of this question. Dogan, Weiss, Patel, Cheong and Gunes ran image-to-text-to-image loops on activity and emotion datasets and found that the demographic distribution of the pictured people drifts toward younger and more female-presenting, with some predictions resting on spurious cues such as hair and background (Dogan et al., 2026). Their design measures drift in a distribution. The design here seeds controlled attributes and measures their survival with hazards, against a one-shot baseline that says which prior is pulling. The two designs disagree on the direction of gender drift, and a later section takes that up.

The experiment is a pilot. It was sized to be run on a rented GPU for under forty dollars, and its purpose was to find out whether the question had a measurable answer before asking anyone with expertise to invest in it.

What was run

Loop. Stable Diffusion XL base and LLaVA 7B, the pair Hintze used, with a caption instruction that asks for a plain photographic description. Each trajectory runs 20 steps: render, describe, render the description. Caption temperature 0.9.

Seeds. 36 occupation prompts: airline pilot, nurse and social worker, each crossed with young or older, man or woman, and light, medium or dark skin, written as "a photograph of a young dark-skinned woman working as a nurse" and similar. The three occupations were chosen for their one-shot stereotypes: pilots render as light-skinned men, nurses as light-skinned women, social workers as women with dark or medium skin. Twelve portrait prompts with the same attributes and no occupation are the control. Four trajectories per seed with different generator seeds, 192 trajectories per arm, 3,840 images per arm.

Two arms. The Photo arm uses the photographic instruction alone. The Tracking arm adds one sentence: "State the apparent age, gender and skin tone of any person." The Tracking arm tests whether the loss is the captioner failing to mention the attribute or the generator failing to render it.

Judge. Every image was coded by Claude Sonnet 5.5 for person present, number of people, and the primary person's gender presentation, age band, skin tone band, face visibility and framing, with Claude Haiku 5.5 as a second opinion. I labelled 150 images by hand before the runs to calibrate the judge. Agreement with my labels:

field	Sonnet kappa	Haiku kappa	Sonnet and Haiku kappa on the full set
gender	0.83	0.87	0.84 to 0.89
age	0.76	0.71	0.63 to 0.86
skin tone	0.71	0.70	0.73 to 0.81

My disagreements with the judges were directional: I placed the young-to-middle boundary later and every skin-tone boundary one band darker. The judge is therefore conservative about calling a face dark, which matters for reading the skin results.

Skin tone by measurement. Because judge bands are coarse, every face was also measured: a face detector, a skin segmenter, and the mean CIELAB lightness L* of the skin pixels, higher meaning lighter. This is the robust readout for skin and the judge band is the secondary one.

One-shot baseline. 450 images rendered once from each occupation prompt and from the neutral "a photograph of a person", judged the same way. This says what the generator's prior is for each occupation and for a person with no occupation, so that the loop's end state can be compared with both.

Survival. An attribute counts as kept at step k when the judge's reading of the primary person matches the step-0 reading. Loss is measured relative to what the generator actually rendered at step 0, not the prompt, because SDXL rendered "light-skinned" as medium in half the seeds and "medium-skinned" as dark in a third. Dropout of the person is treated as a competing risk. Each loss is classified by mechanism from the caption that produced the image: omission (caption named no value), caption change (caption named a different value) or override (caption named the seed value and the generator ignored it).

Result 1: seeded demographics are lost within a few steps

Skin tone goes first and goes at once. Gender and age follow over about ten steps. No occupation trajectory in either arm still showed all three seeded attributes at step 10, and by the end of the run most trajectories no longer showed a single person.

Share of occupation trajectories still showing the seeded value, 144 per arm (portrait control in brackets, 48 per arm):

attribute	arm	kept at step 5	kept at step 10	kept at step 19
gender	Photo	0.47 (0.83)	0.23 (0.63)	0.09 (0.25)
gender	Tracking	0.59 (0.83)	0.34 (0.54)	0.13 (0.31)
age	Photo	0.23 (0.54)	0.08 (0.21)	0.01 (0.06)
age	Tracking	0.36 (0.69)	0.12 (0.46)	0.02 (0.23)
skin band	Photo	0.05 (0.13)	0.03 (0.08)	0.01 (0.04)
skin band	Tracking	0.10 (0.10)	0.07 (0.08)	0.03 (0.06)

The measured lightness tells the skin story without the judge. Mean facial L* of seeds by their seeded band, Photo arm:

seeded band	step 0	step 1	step 2	step 5	step 10	step 19
dark	24.9	50.1	53.5	49.9	49.7	50.2

seeded band	step 0	step 1	step 2	step 5	step 10	step 19
medium	41.4	50.1	54.3	52.8	54.0	51.1
light	48.1	54.4	52.0	53.2	54.0	53.1

One step is enough. Every dark-skinned seed, 64 per arm, had left the dark band by step 5 in both arms. The three seeded bands sit on one distribution from step 2 onward and stay there. Light seeds keep their band, with a loss hazard of 0.02 to 0.11 per step, so the loss runs in one direction, toward light, for everyone.

This is the homogenisation that Wilson, Ghosh and Caliskan measured in SDXL for a single prompt, applied twenty times in a row. A loop is a homogenisation amplifier: whatever variability the generator removes per render, it removes again at every step, and the captioner gives none of it back.

Gender loss is directional too. Counting changes from a kept state in the Photo arm, women became men 24 times and men became women 7 times among pilots; 31 and 11 among social workers; 20 and 10 among portraits. Only nurses balanced, 42 against 36. The social worker, whose one-shot prior is 97% women, lost women to men three to one.

The portrait control loses more slowly on every attribute, which says the occupation accelerates the loss, but it reaches the same place: 4% of portraits kept all three attributes at step 10, and 0% were still a single person in head-and-shoulders framing at step 19, from 83% at step 0. Being a person with no story is not protection.

Result 2: the replacement is the loop's prior, not the occupation's

The prediction written before the run was that a lost attribute would be replaced by the occupation′s stereotype: dark-skinned pilots would turn light because pilots are rendered light, and dark-skinned social workers would be protected because social workers are rendered dark. The first half held and the second did not. The loop pulls every seed toward the same place, and that place is the generator′s rendering of a person with no occupation at all.

The one-shot baseline gives the three candidate priors. Rendered once from the bare occupation prompt, with the judge′s reading and the measured facial lightness:

one-shot prompt	men	light skin	dark skin	mean facial L*
airline pilot	89%	86%		39.3

one-shot prompt	men	light skin	dark skin	mean facial L*
nurse	1%	81%		41.3
social worker	3%	12%	50%	42.1
"a person" (no occupation)	56%	54%	4%	55.5

The loop's step-19 faces have mean L* between 52 and 58 in every occupation and both arms. That is the neutral prompt's value. It is lighter than the one-shot faces of every occupation including the pilot, whose workplace lighting keeps even light-skinned faces at around 39. The loop does not drift toward pilots; it drifts past them.

The hazard comparison makes the same point. If the occupation's prior were doing the pulling, dark-skinned social workers should lose their skin tone more slowly than dark-skinned pilots. Per-step hazard of leaving the dark band:

seed	Photo arm	Tracking arm
dark-skinned pilot (one-shot prior: light)	0.70	0.52
dark-skinned social worker (one-shot prior: dark)	0.82	0.58

There is no protection. For gender the counter-stereotypical and stereotypical seeds lose at the same hazard, ratio 0.9 to 1.0, and the direction of the change is toward men in every occupation except nurse. The share of all changes that land on the occupation's one-shot value is 36% in Photo and 41% in Tracking, against the 70% that was predicted.

The reason, worked out in the next section, is that the occupation leaves the caption before the attribute does, and once it is gone the loop has no occupation to be stereotyped about. What is left is the generator's and captioner's shared default for a person, which in this pair is light and male. The practical implication for bias measurement is that the one-shot stereotype, the thing most audits measure, is the wrong reference for what a loop will do. The quantity that matters is the loop's own stationary prior.

Result 3: the captioner is the channel

The loop loses attributes mainly because the captioner stops mentioning them, and when it does mention skin tone it often gets it wrong in one direction. The occupation itself is subject to the same filter: one of the three seeded occupations is never named in any caption of either arm.

Omission. In the Photo arm, 87% of attribute losses (823 of 942) happened at an image whose caption named no value for that attribute at all. The generator was not overriding the caption; it was filling a blank. The caption's naming of the attributes fell over the run: gender was named in 81% of step-0 captions and 29% at step 19; age 47% to 12%; skin 14% to 2%. The attribute drops out of the words first and out of the pictures a step later.

Lightening in the words. Under the Tracking instruction the captioner names a skin tone most of the time, 77% at step 0 and still 28% at step 19, yet skin survival barely improves. Of 113 losses of a seeded dark or medium band in that arm, 52 followed a caption that named a different band, almost always "skin tone is fair" or "light" for a face that the judge and the lightness measurement both read as dark or medium. The captioner lightens the person before the generator does. The generator then adds its own floor: in the check set it rendered "light-skinned" and "medium-skinned" as the same overlapping distribution, L* 47 against 42, so even a correct caption cannot hold a medium band.

Invisible occupations. The word "pilot" appears in 85% of step-0 captions and "nurse" in 48%. "Social worker" appears in 0% of captions at every step of both arms. From step 0 the social worker seed is "a person at a desk", and the social worker's one-shot prior, 97% women and half dark-skinned, has no channel into the loop. The job or its setting is still in 65% of pilot captions at step 18, 50% of nurse captions and 60% of social-worker captions, but for the social worker the setting is an office, which carries no demographic prior at all.

The three mechanisms rank the same way across the data: omission is the plurality in both arms (570 of 1,063 losses even under Tracking), caption change second, generator override last. A loop's bias on demographic attributes is, in this pair, mostly a bias about what is worth saying.

Result 4: asking the captioner to track attributes helps some of them and creates a new attractor

One added sentence, "State the apparent age, gender and skin tone of any person", doubled how long gender and age survived, did nothing for skin, and turned more than half the images into grids of faces by the end of the run. A measurement instruction is not a passive probe.

What it helped. Gender kept at step 10 rose from 23% to 34% of occupation trajectories, age from 8% to 12%, and the person stayed in frame far longer: 91% of occupation trajectories still showed a person at step 19 against 64% under Photo. Omissions fell by

31%, from 823 to 570, which was less than the 50% predicted, and they remained the largest mechanism.

What it did not help. Skin kept at step 5 moved from 5% to 10%. The two reasons are in Result 3: the captioner names dark faces "fair", and the generator cannot hold a medium band on request. Telling a captioner to say the skin tone does not help when the captioner says the wrong one.

What it created. Once two or more people are in frame, the tracking sentence turns into the phrase "individuals of various ages, genders and skin tones". That phrase appears in 0% of Photo captions and rises from 2% of Tracking captions at step 2 to 17% at step 18. The generator renders it as a collage of dozens of faces. Images with five or more people reach 55% of the Tracking arm at step 19 against 38% under Photo, and single-person images fall to 15% to 23% in both arms from 78% to 98% at step 0. The tracking arm also drifts toward black-and-white and sepia, 38% of images at step 19 against 16%, and one joint cluster, vintage group photographs of uniformed men, is 91% Tracking.

The collage is a new attractor in the sense of the companion draft: a state the loop enters from many starting points and stays in. It did not exist before the instruction. For bias measurement the lesson is uncomfortable. The instruction that makes the attributes visible to the captioner also changes what the loop produces, so a tracking arm needs its own attractor check against a control before its survival numbers mean anything.

Result 5: the loop re-assigns occupations, and that is where the stereotype shows

The captions do not only drop attributes; they promote, demote and re-enlist people. A nurse becomes a doctor when the picture has drifted to a light-skinned man. A woman pilot becomes a male astronaut. A pilot becomes a soldier of either gender. These were not pre-registered; they came from reading the captions after the fact, and they are the clearest stereotype in the data.

Nurse to doctor. Of 48 nurse trajectories per arm, 39 in Photo and 37 in Tracking have a caption that calls someone a doctor, physician or surgeon, typically by step 3. Whether a trajectory ever gets the word does not depend on its seed: male and female seeds and all three skin bands receive it at similar rates. What decides the word is who is in the picture when the captioner writes it. On single-person images from nurse seeds in the Tracking arm, where captions are gendered:

caption calls the person	judge sees a man	judge sees a woman
doctor	18	4
nurse	1	13
all single-person images	87	87

Fisher's exact test $p < 0.001$. The doctor images are light-skinned in 20 of 22 cases; the nurse images include 4 dark and 2 medium of 15. A male nurse seed is called a doctor at a median of step 2 while the picture still shows a man. A female nurse seed is called a doctor at a median of step 4, by which time the picture shows a man in roughly a third of cases. The sequence is: the generator drifts the nurse toward a light-skinned man, then the captioner re-labels that man as a doctor and keeps "nurse" for the women who remain. The Photo arm points the same way, 12 men and 2 women for doctor, with too few gendered captions to be significant on its own.

Pilot to soldier is not biased; pilot to astronaut is. 42 and 44 of 48 pilot trajectories become military, and under Tracking the military word is in 54% of captions at step 19 while "pilot" has fallen to 12%. This conversion carries gender and skin through unchanged: military captions describe 73 men and 88 women in Tracking, the same split as pilot captions and as all single-person pilot images, and the skin bands match the baseline. The generator renders women in uniform as readily as men, so nothing is lost. Astronaut captions describe 35 men and 11 women in Photo and 22 men and 0 women in Tracking, $p = 0.003$ and $p < 0.001$ against the military split, and in Photo 11 of the 18 astronaut trajectories started from woman seeds.

Social workers become nothing. No job word ever appears for them beyond "office". The one occupation seeded for its counter-stereotypical prior is the one the loop cannot see.

What I take from this, and would like checked by someone who studies occupational stereotypes: the attribute loss in Results 1 to 3 is a bias of attention, but the re-labelling is a bias of association. The captioner has a lexicon in which a light-skinned man in scrubs is a doctor and a woman in scrubs is a nurse, and the loop gives that lexicon many chances to apply.

Result 6: whose defaults

The light-and-male default belongs to this pair, not to loops in general. Across the other runs in the project, which were not designed as bias experiments but kept every caption, the gendered vocabulary a loop ends with depends on the captioner, people leave the pictures under every captioner, and the language of the instruction decides the ethnicity of the faces.

loop (100 trajectories, 100 steps, Hintze's seeds)	male to female words, last quarter	captions naming a person, last quarter	images with a face, step 0 to step 99
SDXL + LLaVA (baseline)	298 : 183	22%	58 to 16
SDXL + Qwen2.5-VL	0 : 39	14%	
SDXL + Claude Sonnet 5.5	64 : 518	17%	58 to 10
Hunyuan + Qwen2.5-VL, English instruction		53%	53 at step 99
Hunyuan + Qwen2.5-VL, Chinese instruction	31 : 260	22%	29 at step 99

Three readings:

- **The direction of gender drift is the captioner's.** LLaVA's loops end male, Qwen's and Sonnet's end female by large ratios. Dogan and colleagues found drift toward female-presenting people in their loop. The disagreement between their result and the E6 result is probably not a disagreement about loops but about which captioner was in them, which is testable.
- **People drain out of the pictures under every captioner.** Faces fall from 58 of 100 images to 10 or 16 by step 99 in the SDXL cells, and Sonnet, the strongest captioner, drops them fastest. A loop's long-run default is not a particular kind of person; it is no person, with interiors and landscapes in their place. The demographic bias is a property of the transient, which in E6 is the only part measured.
- **The language of the instruction sets the ethnicity.** With the caption instruction in Simplified Chinese and the same English seeds, faces became East Asian within one to five steps, 15 of 29 faces in the first five steps against 0 of 29 under English, with

blonde seeds exempt. The judge declined to code apparent ethnicity from faces and was recorded verbatim; the counts are an eye-shape proxy and my own count, and a native-speaker review of the Chinese captions is still pending.

None of these runs used demographic seeds, a judge on every image or a one-shot baseline, so they are context for E6 rather than results of the same standard. They are the reason the next experiment is a captioner swap.

Comparison with Dogan et al. (2026)

The two studies ask the same question with opposite designs and get opposite gender directions, and the opposition is probably informative rather than contradictory. This comparison rests on their abstract; I have not yet worked through the full paper and will revise this section when I have.

	Dogan, Weiss, Patel, Cheong and Gunes (2026)	this draft (E6)
starting point	real photographs from activity and emotion datasets (PHASE, RAF-DB)	generated images from 36 controlled attribute-by-occupation prompts
what is measured	drift of the demographic distribution across the loop	per-trajectory survival of a seeded attribute, hazard per step, mechanism of each loss
reference	the starting dataset′s distribution	the generator′s one-shot prior for the occupation and for a neutral person
gender direction found	toward female-presenting, mainly for emotions	toward male, in every occupation but nurse
age direction found	toward younger	both ways within occupation; no directional effect
extra readouts	explainability: predictions rest on spurious cues such as hair and background	re-labelling of occupation conditioned on the rendered person

Why the gender direction may differ. Result 6 shows the direction of gender drift is set by the captioner: LLaVA's loops end male and Qwen's and Sonnet's end female. If their loop used a captioner other than LLaVA, the two results agree on the mechanism and disagree only on the default. The test is cheap: run their datasets through this pair and these seeds through theirs.

Why the age direction may differ. Their drift toward younger is measured on real faces with a full age range. E6 seeds only "young" and "older", and the judge's young-to-middle boundary is earlier than mine, so an age drift toward the judge's "young" would be hard to see here. Their finding is probably the better one on age.

What each design cannot see. A distribution measure cannot tell omission from override, cannot say whether a dark-skinned person was replaced by the occupation's stereotype or by a generic default, and cannot see re-labelling. A seeded-survival measure cannot say what happens to a realistic population, is limited to the few occupations seeded, and depends on a judge calibrated by one person. The two together would be a much better paper than either, which is the main reason I am circulating this draft.

Next experiments

Two experiments are defined well enough to run now, and a third needs an expert to design. Each will have a written prediction before it starts, as every run so far has.

E6-C: a frontier captioner. The same 192 seeds, 20 steps and 4 replicas, Photo and Tracking arms, with Claude Sonnet 5.5 replacing LLaVA and nothing else changed. It separates instruction-following from attribute drift: Sonnet held the 50-word caption limit in 99.5% of 10,000 captions in an earlier run where LLaVA held it in 0.3%, so if attributes still vanish under Sonnet the loss is not a failure to follow the instruction. The earlier run also showed Sonnet drops people from the pictures faster than LLaVA and ends 8:1 female in its gendered words, so the result is not predictable. Cost is about 15 dollars of API per arm plus two hours of GPU. One design question is open: Sonnet is also the judge. For this cell I would make Haiku the primary judge and report Sonnet's agreement, or add a judge from another model family, and I would value advice on which.

E6-O: occupations the captioner can see. The social worker result means one third of the E6 design had no channel. The fix is a legibility pre-test before seeding: render each candidate occupation once, caption it, and keep only occupations the captioner names in at least 80% of step-0 captions. Candidates with visible props or uniforms: chef, police officer, firefighter, construction worker, scientist in a lab coat, flight attendant, surgeon, judge in robes. The run then measures the occupation's own survival alongside the

attributes, and pre-registers the re-labelling matrix (which occupations are promoted, demoted or re-enlisted, conditioned on the rendered gender and skin) that Result 5 found after the fact. Same size as E6, about 40 dollars.

E6-X: the expert's design. This is where I need help, and the questions are specific:

1. Which occupations and which attribute levels make the strongest test of stereotype pull versus generic default? The pilot, nurse and social worker set was chosen by two amateurs from one-shot renders.
2. Is a three-band skin tone the right construct, or should the seeds and the judge use the Monk scale, and does measured facial lightness stand in for it adequately?
3. How should "ambiguous" gender presentation be scored? The pre-registered rule counts it as a loss; the sensitivity analysis with it as unknown changes no verdict, but the choice is a values question.
4. In crowd images, which person's attributes count? E6 codes the primary person and treats the drift to crowds as a competing risk, which hides what the crowds look like.
5. Is there an established way to code apparent ethnicity in synthetic faces that a model judge will apply? Sonnet declined to code it in the Chinese-instruction run, and the eye-shape proxy used there is not a measurement anyone should publish.

A longer-term pair of runs. The Dogan exchange above, their datasets through this pair and these seeds through theirs, would settle whether the direction of drift is the captioner's. And the companion draft's rejection experiment could be re-run with demographic seeds to see whether keeping a loop out of its attractors also keeps the person in the picture.

Limitations

- **One model pair, one 7B captioner, 20 steps, 4 replicas per seed.** Everything in Results 1 to 5 is about SDXL and LLaVA. Result 6 is the only evidence that other pairs behave differently, and it comes from runs not designed for the question.
- **The judge is a language model calibrated by one person.** Agreement with my labels is kappa 0.71 to 0.83, and my own boundaries sit darker and older than the judge's. Skin "kept" by judge band is harsh at the light-to-medium boundary; the measured lightness is the robust skin readout and is the one the conclusions rest on.
- **The generator did not render the seeds faithfully.** "Light-skinned" came out medium in half the seeds and "medium-skinned" came out dark in a third. Survival is measured against what was rendered at step 0, which is the right correction but means the

seeded design is weaker than it looks.

- **One seeded occupation was invisible to the captioner.** The social worker arm tests nothing about social workers. It does, by accident, show what happens when an occupation has no channel, which became Result 2.
- **The mechanism split depends on a keyword list.** Omission, caption change and override are decided by matching attribute words in the caption. Two bugs in the matcher were found and fixed on this data before any number was read, and the skin phrasings LLaVA uses under Tracking were added after inspecting captions. The split is reported as dependent on that list.
- **Result 5 is post hoc.** The re-labelling analysis was done after reading the captions, in answer to a question I asked when the occupation turned out to be less durable than I expected. It should be pre-registered and replicated in E6-O before anyone relies on it.
- **"Ambiguous" gender counts as a loss** by the pre-registered rule. With it counted as unknown, gender kept at step 10 rises from 23% to 31% and the hazard ratios do not move.
- **Claude did the engineering and the analysis, and Claude is one of the captioners and the judge.** I set the questions, confirmed every prediction before each run, labelled the check set and read every report; Claude Code wrote the runner, the judge prompt, the analysis and the first draft of this document. Pre-registration guards against fitting the story to the data; it does not guard against a shared blind spot, and it does not resolve the conflict of a model family judging its own captions, which E6-C has to address.
- **Coding apparent demographics from synthetic faces is itself a contested practice.** The judge codes presentation, not identity, and the images contain no real people, but the categories are still the ones the field argues about. I would rather be told how to do this properly than keep improvising.

Acknowledgement

This project was a collaboration between the author and Claude (Anthropic). Claude was used through Claude Code as engineer, analyst and writing partner, and through the Claude API as one of the captioners under test and as the judge that coded every image. The author set the questions, chose and sized the experiments, wrote or confirmed every prediction before each run, hand-labelled the judge calibration set and read every report. Claude wrote the runner and the analysis code, drafted the quantitative thresholds of the predictions, proposed several of the measurements (the one-shot baseline, the mechanism split, the hazards with confidence intervals), built the reports and wrote the

first draft of this document, which the author then revised. Where Claude's quantified predictions and the author's directional ones differed, both are recorded in the repository with the outcomes. Every number here was computed by code in the repository from the run data. The author takes responsibility for the claims. A separate note describes the collaboration in detail.

Data, code and references

Code and data. All runs, seeds, predictions, judge outputs, face measurements, analysis scripts and reports are in a git repository (github.com/bradhinkel/ai_to_ai_loops, private while this draft circulates; access on request). Every image and caption is kept with model versions, generator seeds and package versions logged per step. The 150-image hand-labelled check set and the judge agreement tables are included. The E6 gate report with every table behind this draft is reports/e6.md in that repository, and the occupation re-assignment tables are reports/e6/occupations.md.

Models. Stable Diffusion XL base 1.0; LLaVA 7B (Ollama llava:latest, Q4_0); judge Claude Sonnet 5.5 with Claude Haiku 5.5 as second opinion; face detection YuNet; skin segmentation CelebAMask SegFormer; lightness in CIELAB. Other loops cited in Result 6 used Qwen2.5-VL 7B, Hunyuan-DiT v1.2 and Claude Sonnet 5.5 as captioner.

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Companion draft. Gravity in AI-to-AI loops: attractors as temporary stops, with radius, depth, and what was tried to keep loops out of them.